Artificial intelligence helps drive new frontiers in ecology

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Feature

Artificial intelligence helps drive new frontiers in ecology

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Monarch butterflies at the University of Florida's McGuire Center. Photograph: Vaughn Shirey.

Vaughn Shirey has spent hundreds of hours in museums photographing butterflies, working among drawers full of flaming orange monarchs and yellow-splashed skippers. Shirey is interested in how global change is affecting high-altitude butterfly biodiversity, but butterflies have been poorly sampled and studied in the past. "We don't have a time machine, so we can never go back to the 1970s and 1980s and conduct a survey," they say. "So our best record is natural history collections."

As part of their dissertation at Georgetown University, in Washington, DC, in the United States, Shirey used the butterfly photos to study the darkly shaded areas on the hindwings of the cold-adapted Rocky Mountain parnassian (*Parnassius smintheus*), an otherwise mostly white butterfly speckled with orange and black dots. Dark areas have more melanin, a pigment thought to absorb solar radiation and direct heat energy to the butterfly's flight muscles. Shirey wanted to know whether increasingly warmer temperatures are changing the melanin patterns on the parnassian's wings over time. Melanin affects flight ability and provides other benefits. Changes to melanin could help or hinder the threatened butterfly's ability to adapt to climate change and habitat loss in the alpine.

A short time ago, Shirey would have had to painstakingly sort through their collection of 2000 butterfly photos, measuring each pixel on each wing related to potential changes in pigment. "It would have been extremely painful," they say, laughing. "Each one would have taken hours." Instead, they opted to use computer vision, a form of machine learning that enables computer artificial intelligence (AI) to derive information from images, videos, and other inputs. The AI processed the photos in a much more consistent and efficient manner than manual approaches. Shirey did not find that the amount of melanin on the hind wings changed over time, but there were some important differences between male and female butterflies that may make females more vulnerable to climate change.

Shirey, now a postdoctoral conservation fellow at the University of Southern California's Washington, DC, campus, is one of many scientists harnessing the power of AI and machine learning to help analyze vast amounts of data and to monitor changes to species and ecosystems. AI is helping ecologists monitor air quality, measure the changing footprints of ecosystems, and track changes to species distribution. Beyond facilitating the ability to collect, connect, and analyze data at unprecedented scales, AI is allowing scientists to build predictive models with the potential to revolutionize ecology and the environmental sciences parallel to the way statistics transformed these disciplines in the twentieth century. AI may prove invaluable in navigating complex ecosystem issues such as climate change, biodiversity loss, and the spread of zoonotic diseases. However, there are many considerations in its use. AI is only as effective as the information it is fed. Data access, quality, and privacy are known challenges. Training, infrastructure, and cost are also barriers to widespread use. And there are considerations around ethics and governance; a large proportion of AI tech is developed in the private sector, which is, by necessity, for profit. As AI becomes an increasing presence in the way ecologists seek to understand the world, the way forward may require their active participation in how the technology unfolds.

The rise of AI in ecology

Humans have long been fascinated with building intelligent machines. Several decades before ChatGPT arrived on the scene, AI pioneers were already using the term *artificial intelligence*. In 1950, scientist Alan Turing developed a test to determine whether a computer could think like a human, later coined the Turing test. In 1985, IBM began to develop an intelligent chess computer called Deep Blue. It beat world chess champion Garry Kasparov in 1997. Deep Blue was a symbolic AI, one limited to systems that operate by well-defined sets of rules, writes Soha Hassoun and colleagues in their article "Artificial Intelligence for Biology"

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Vaughn Shirey says the Rocky Mountain parnassian (*Parnassius smintheus*) is on "the elevator to extinction" as climate change causes forests to move up mountainsides and overtake the butterfly's meadow habitat. AI helped Shirey understand butterfly changes in response to heat. Photograph: Vaughn Shirey, University of Florida—McGuire Center.

(http://doi.org/10.1093/icb/icab188). It was a powerful step forward, but the living, breathing biological world is much messier. Symbolic AI can only make choices based on an established set of rules. "Biological intelligence, however, can learn on the fly and make decisions based on information acquired by experience and by seeing objects, for example," explains Hassoun.

The AI ecologists use today is much more than symbolic. Harnessing the power of machine learning, AI can learn and adapt without following explicit instructions. It can process and analyze vast volumes of data more efficiently, make predictions, and potentially gain insights into ecological systems that would be challenging with traditional statistical analyses alone. AI can learn in a nonlinear way. This gives it the potential to cut through a lot of human biases in data collection, as well as our assumptions about the data, says Vaughn Shirey. Often, statistical models are quite linear. "A lot of machine learning approaches don't have the same assumptions as those models," they say, which may make AI better able to make sense of complex biological systems.

Understanding the natural world with AI

Whether collecting lilting bird song in audio recordings or snapping photographs of elephants, wolves, and other wildlife with camera traps, ecologists are generating more data about the natural world than ever before. Processing and analyzing the volumes of these data, however, has been daunting and time consuming. Helping scientists make sense of information collected from drone footage, camera traps, satellites, microphones, and community science platforms such as iNaturalist is one big area of promise for AI.



A representative of the skipper family (Hesperiidae), which Vaughn Shirey is now working on from Yale Peabody Museum. Photograph: Vaughn Shirey.

While Shirey was a PhD student, they and colleagues made use of AI to classify satellite imagery to determine the extent to which a flood-tolerant saltmarsh plant, *Spartina alterniflora*, was replacing the flood-sensitive *Spartina patens* as a result of sea-level rise. These plants are foundational to the expansive, grassy saltmarsh ecosystems that line the estuarine coastlines of many eastern US states. Intertidal habitats like these are nurseries for many of the east coast fisheries and are important for carbon sequestration, storm buffering, and pollutant filtering. They are quickly deteriorating as sea levels rise, the waters become polluted, and the coastlines are developed. Although scientists have been able to monitor the loss of saltmarshes at a landscape level, there is a need to understand fine-scale effects on species, too, so that appropriate mitigations can be assessed.

The team trained the AI to detect characteristics of each plant, especially color, and examined two similar-size marshes from the same area in New Jersey. As a result, they learned that one of the marshes experienced very little change between 2006 and 2019, whereas, at the same time, the other experienced an over 80% loss of *S. patens*. The findings prompted a closer look. They discovered that the loss of *S. patens* correlates with increased streamflow and total nitrogen content in rivers running through each marsh. "One of the promises with this AI method is that we can rapidly detect turnover like that and pinpoint the cause very accurately," explains Shirey.

Similar technology to what Shirey and colleagues used for their regional marsh study can be scaled up to the national or even international level. Global Forest Watch

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(www.globalforestwatch.org) is an online platform administered by the World Resources Institute (WRI) that provides data and tools for monitoring forest ecosystems and levels of forest cover. Since 2015, the program has used computer vision and machine learning to monitor activities related to deforestation, such as the internationally extensive development of oil palm plantations. It can be hard to tell palm tree plantations from forests on satellite images, but WRI is training AIs to tell the difference. In the long term, they hope to be able to better detect and understand the emergence of the plantations and forest conversion. The work is supported by the United Nations Environment Programme (UNEP), a partner in the program's development. Monitoring large-scale ecological change with AI is a major area of interest for the United Nations, says David Jenson, coordinator of the UNEP Digital Transformation Programme. In 2022, member states asked the UNEP to examine how digital technologies like AI could accelerate work in three areas: climate action, nature protection, and pollution prevention. In response, the UNEP launched the World Environment Situation Room (wesr.unep.org), a digital platform that is planning to leverage AI's capabilities to analyze complex, multifaceted data sets. The hope is that, over time, the Situation Room will monitor a suite of environmental indicators that can drive environmental action at different scales.



Coordinator of the UNEP Digital Transformation Programme, David Jensen says data quality is an issue when it comes to making datasets that can be used for global ecosystem monitoring via AI. In response, the UNEP is developing a Global Environmental Data Strategy. Photograph: Matija Potocnik.

It is not just about using AI to analyze new data sets. "I've always been very interested in how to do more with the data we have already collected," says Sara Beery, an assistant professor in the faculty of AI and decision making at MIT (in Cambridge, Massachusetts). She helped develop Auto Arborist (https://google.github.io/auto-arborist), an urban tree classification data set made from a merger of available tree censuses and data collected through Google's Street View and satellite imagery. Most cities on Earth have Street View data, says Beery. "That data was all paid for and collected [by Google] because people wanted to be able to map cities better." But the data can be used for so much more. Beery used machine learning to train an AI to differentiate between trees species. Now, Auto Arborist identifies trees in 23 cities in the United States and Canada, and work is ongoing to



MIT's Sara Beery is interested in how machine learning such as computer vision can help extract more information from existing datasets. Photograph: MIT EECS.

expand this to track the size, health, and species distribution of trees over time. The goal is affordable urban forest monitoring at scale. This data can help city policymakers, city planners, and researchers quantify the habitat and ecosystem services provided by urban forests, such as carbon sequestration, improvements to air quality, and other public health benefits. It can also help track damage from extreme weather events and help target tree planting to improve each city's resilience to climate change, as well as diseases and insect infestations.

AI is useful not just for monitoring changes to the natural world but potentially for predicting change as well. For example, AI can help understand the complexity around phenomena such as wildfires and can help make predictions about potential risks, says Jensen. Barbara Han is a disease ecologist at the Cary Institute of Ecosystem Studies, in Millbrook, New York. For her, prediction is a critical aspect of her work. "We're really interested in figuring out which animal species are going to give rise to zoonotic diseases," she explains. Han and her colleagues are investigating whether machine learning can differentiate animal disease carriers from noncarriers. In a 2019 study, they looked at whether machine learning could help predict which bat species harbor filoviruses such as Ebola. That's important work, given the devastating effects of the 2014 Ebola outbreak in West Africa on people and primates. "When the Ebola outbreak happened, there [was] not a clear understanding of which species may be serving as reservoirs," says Han, adding that this is still a work in progress today.

Using machine learning, the team trained an AI to recognize the traits of bats that have tested positive for filovirus antibodies, such as their life histories, movement, and diet. Then they used the AI to search a database of the world's 1116 known bat species to identify potential new hosts. Later, in field studies, some of the new hosts identified by AI were confirmed to be carriers of filovirus antibodies. When Han and her team mapped all the hosts the AI identified, they found the bats more widely distributed than

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expected. In a press release, Han said, "Our results corroborate studies that have predicted the environmental niche of Ebola to span primary tropical rainforest in Africa. But in a departure from past research, we identified several hotspots of filovirus-positive bats in Southeast Asia, where up to 26 species overlap, notably in Thailand, Burma, Malaysia, Vietnam, and northeast India."



At the Cary Institute of Ecosystem Studies in Millbrook, New York, Barbara Han is using AI to help predict which animals will give rise to zoonotic diseases. Photograph: Cary Institute of Ecosystem Studies.

Han is now using machine learning to colead a study on how environmental conditions shape viral outbreaks in rodents. Like her work on Ebola, the aim is to minimize spillover to people. Similarly, she continues to use AI to predict interactions among humans, wildlife, and SARS-COV-2 strains that not only spill over to humans but could also to move from humans to wildlife. In a recent paper, Han and her coauthors used machine learning to predict which of 5000 wildlife and domestic species are most susceptible to SARS-CoV-2. Cougars, white-tailed deer, domestic cats, and 18 other animals emerged high on the list. Han says this work could aid wildlife managers responsible for the health of whole ecosystems. Useful information might include, for example, how diseases such as SARS-CoV-2 can upset predator–prey dynamics. Han's future work will employ machine learning to tackle these complex interactions.

There is little doubt that machine learning and other AI tools will continue to revolutionize the fields of ecology and environmental science by allowing scientists to analyze large volumes of data, to detect patterns otherwise hard to see, and to help make predictions in complex ecological systems. However, as Han says, all this work is only as good as the data going into the analyses.

Garbage in, garbage out: Data management

Machine learning and AI are notorious for needing lots and lots of training data, says Han, who has spent many hours poring over natural history papers and other sources to produce data sets. It is not just about the volume of data; it is about the quality. "Machine learning can only make predictions based on the data that comes in," says Melissa Guzman, lead of the Ecological Science Lab at University of Southern California Dornsife, Los Angeles. It is not uncommon to have biases in data sets. She points to community collected data sets such as iNaturalist, where many of the observations are collected near cities and roads because that is where people are concentrated. Data are also more likely to be collected during the middle of the day because that is when humans like to sample, she says. "There's all these nighttime organisms that get forgotten."

In addition to data biases, sometimes data is misrepresented, mislabeled, or imperfect in some way. For example, when a citizen scientist is trying to identify an insect such as a beetle, they can miss subtle differences that separate species simply because they have not been trained in identification. But when scientists want to work with data from community platforms such as iNaturalist, the Global Biodiversity Information Facility, and others, manually cleaning data sets of these errors is incredibly time consuming. Guzman and postdoctoral researcher Jayme Lewthwaite are investigating how machine learning can be trained to identify these erroneous data and clean up data sets, using butterflies as their subjects. So Guzman is using machine learning not only to analyze data but also to clean the data that needs to be analyzed.

Data quality is one of the biggest challenges for the United Nations right now, along with data interoperability (the ability of different computer systems or software to exchange and make use of information), says David Jensen. When countries have vastly different methods of collecting and classifying data, it is challenging for the United Nations to aggregate national data into global data sets for analyses. As a result, "We have a very hard time monitoring the level of compliance with the different environmental conventions," he explains. "It's one of the reasons that UNEP has been asked to develop a Global Environmental Data Strategy by 2025. The data strategy aims to enhance international engagement with new technologies to improve data standards and digital cooperation between countries.

Data shortages and access can also be problematic. If Barbara Han wants to investigate which animals host avian influenza and where the virus is more likely to spillover to human populations across the globe, it takes enormous effort to find that information, she says. The task requires pulling data on global pollution, biodiversity, and agriculture. "So now you're crossing disciplinary barriers in terms of the data," she explains. "And you're not just looking at ecology or surveillance data, which is a little bit more standardized. But you're asking for data from environmental epidemiology and the agriculture industry. And there's no incentive to share these data." In other words, many data are proprietary. Accessing them can require copious time and resources, if they are available at all.

Even if there is access to data, extreme care needs to be taken to ensure data are managed respectfully, without causing harm. For example, "If you work with endangered species, you have to be really careful about identifying them," says Guzman. "You don't want to drive further extinction by letting everyone know where they are." And when using aggregated data from multiple sources, it is important that the data collectors are properly acknowledged. There is also the question of privacy. For example, camera traps

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MIT's Sara Beery and team have developed a platform called ElephantBook that, in part, uses computer vision to identify individual elephants and monitor the at-risk elephant population in the Greater Mara Ecosystem, in Kenya. Photograph: Chags Photography for the Mara Elephant Project.



Melissa Guzman leads the Ecological Data Science Lab at University of Southern California Dornsife, Los Angeles, where she is interested in training AI to identify mislabeled data. Photograph: Luigi Manna.

and audio recordings provide extremely useful data for ecologists, but can also capture photos and sounds that people do not consent to, says Vaughn Shirey. "Those are pressing questions from the AI side that we need to think about as we're ramping up some of these tools," they explain. "So much of AI ethics comes down to data ethics."

Forward: Overcoming barriers

Data management is a major challenge of machine learning and AI tools, but there are other considerations too. Many of these technologies are developed by a handful of companies, some of which have become wealthy using machine learning and AI to optimize consumerism. "What AI has done very, very well is helped companies sell more stuff," says Beery. Many of these companies have directed a lot of energy into very specific topics, questions, and data types. "We haven't had as much diversity of application or diversity of topic as maybe we could have." Still, Beery encourages partnerships with these companies where there is potential to benefit from the resources the industry has developed. Her work at Google with Auto Arborist is a good example. She suggests that, in working with ecologists, companies also learn about some of these issues, with potential societal benefits.

"Having the private sector at the table and having dialogue with private sector colleagues on this is essential," says Jensen. "There's this gap between the pace the technology is evolving, and the pace at which we can govern it at a global scale." Jensen says the UN is trying to narrow that gap through initiatives such as the Global Digital Compact, an initiative of the UN Secretary General to create an "open, free, and secure digital future for all," according to the May 2023 brief. The intent is a public–private collaboration with shared objectives.

Removing the barriers for ecologists wanting to use AI is also essential. Funding and infrastructure can be challenges. The systems required to run powerful AI can be expensive, and not all ecologists will have access to the needed infrastructure, suggests Shirey. One big barrier is training. "I think one of the most impactful things going forward is going to be capacity building in the ecological discipline around in house AI skills," says Beery. She codeveloped a workshop called CV4Ecology that helps train students and ecologists how to use machine learning to assist with their research. Currently, there is a lot of reliance on computer scientists to help with these initiatives, says Beery. Learning the basics and understanding the limitation of ready-made AI products and tools will help make conversations more productive, she suggests.

AI and machine learning will likely have broad and lasting impacts on the field of ecology by equipping ecologists with powerful tools to tackle complex questions about the natural world. Fostering the use of AI tools will require attention to overcoming barriers related to data ownership, cost, infrastructure, and training. It will also require finding out where the hype about AI touches down

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Sara Beery captured during camera set up at Mpala Research Centre in Laikipia, Kenya. The cameras were placed to explore the impact of spatial sampling strategies on machine learning performance and animal population estimation. Photograph: Sara Beery and the Laikipia Zebra Project.

in reality, says Jensen. As a member of the UNEP, that means really understanding how these tools will help move the needle on conservation and policy and benefit ecosystems and communities on the ground. For practicing ecologists, it might mean not getting caught up in using AI tools for the sake of using AI tools. "Really consider what matters to you and what works for the questions that you want to ask," suggests Shirey. "Don't forget the ecology."

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